

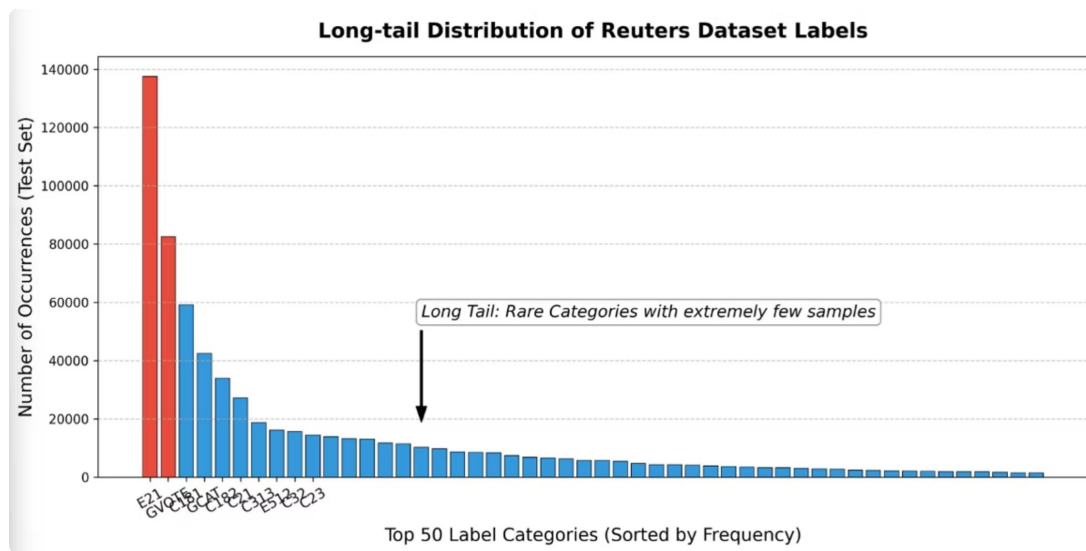


Optimizing Multi-Label Text Classification

A Deep Learning Approach on the Reuters Corpus

Linyao & Aojie

The Challenge & Data Pipeline



Core Challenge

Extreme Class Imbalance: Dominant labels (E21- GOVERNMENT FINANCE/GVOTE - ELECTIONS) vs. Long-tail rare categories.

Data Pipeline

- **Custom XML Parser:** Cleaned 299k docs.
- **Training Sample:** 50k records for faster iteration.

Baseline Methodology

Traditional ML (Control Group)



Logistic Regression

OneVsRest with TF-IDF (Top 20k words).

Deep Learning (Proposed)

DistilBERT

40% fewer parameters; retains 97% performance.

Loss Function

BCEWithLogitsLoss — Industry standard for Multi-label.

Optimization Strategy

Strategy 1: Threshold Sweeping

Scanned 0.30–0.55 to balance Precision & Recall.

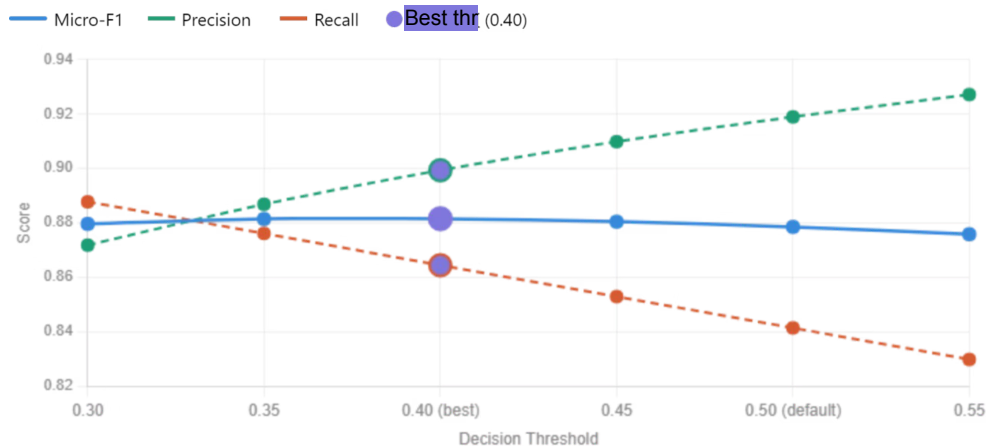
Strategy 2: Stage-2 Fine-Tuning

"Annealing" at $lr=1e-5$ to break convergence plateaus.

Strategy 3: Context Expansion

Increased *max_length* from 256 to 512 to capture terminal clues.
Optimized Batch Size (8) to prevent GPU OOM.

Baseline & Threshold Optimization



Performance Comparison

Baseline (TF-IDF + LR): F1 = 0.8064

DistilBERT (Default thr=0.5): F1 = 0.8785

The Bottleneck

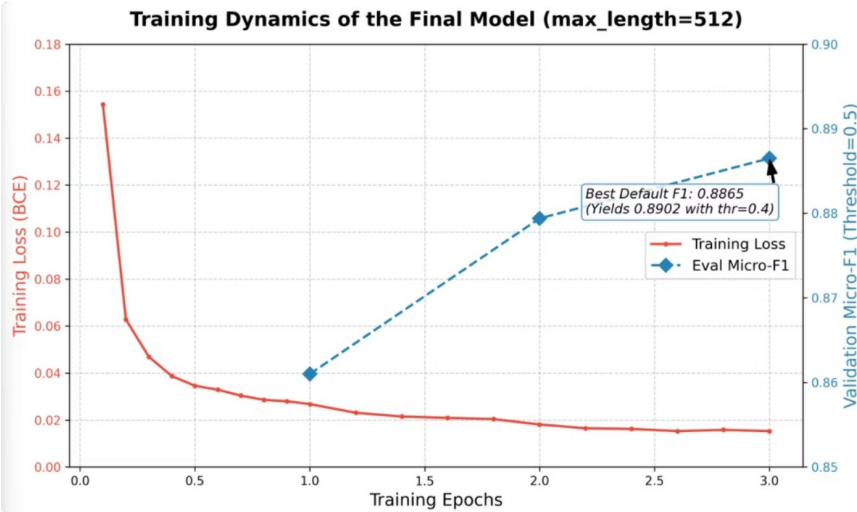
High Precision (91.9%) but Low Recall (84.1%) due to conservative default threshold.

The Solution: Threshold Shift (0.5 → 0.4)

- Significant Recall surge to **0.8644**
- Optimized F1 reached **0.8815**.

Ablation Study: Pushing the Limits

Model	Config	Threshold	F1
Baseline	TF-IDF + LR	0.6	0.8064
Model 1	DistilBERT (50k, 3 epochs, lr=3e-5)	0.5	0.8785
Model 1 (Optimize)	Threshold sweeping	0.4	0.8815
Model 2	Stage-2 fine-tune (lr=1e-5, 1 epoch)	0.4	0.8847
Model 3 (Final)	DistilBERT (512 length, batch=8)	0.4	0.8902



Final Achievement

Model 3 (512 Length) reached peak Validation F1 of **0.8902**.

Key Insight

Breaking the 256-token truncation limit was the most effective boost for long news documents.

Final Blind Test Results

0.8819

Micro-F1 Score

Final Hyperparameters

*max_length=512 | lr=3e-5 | batch=8 |
epochs=3 | threshold=0.40*

0.8747

Precision

Performance Stability

Generalization Gap: Less than 1% difference between Validation (0.8902) and Blind Test (0.8819).

Robustness: Demonstrates zero overfitting despite increased model complexity (512 length).

0.8892

Recall

Strategic Outcome

The Recall-first strategy (thr=0.40) successfully mitigated the long-tail issue, capturing rare labels that a default model would have ignored.

Error Analysis: Semantic Overlap & Model Bias

CASE STUDY: BRE-X MINERALS (LINE 1667)

"...A flurry of news releases, **shareholder lawsuits** and a deepening wrangle between Bre-X and **TSE regulators**... Despite strong wording contained in four **class action lawsuits** filed in the United States..."

- Ground Truth: C11 (Strategy/Plans)
- Prediction: C11 (Strategy) + C12 (Legal/Judicial)



Systematic Finding: the first 5 errors we examined all showed the same pattern — C11 + C12 co-prediction triggered by legal background language.



Key Takeaway: The model is "distracted" by local legal keywords even when they are part of a strategic narrative.

WHY THE MODEL FAILS

Feature Over-indexing

The model gives too much weight to "heavy" legal terms (e.g., *lawsuit*). It treats these local features as primary topics instead of secondary context.

Spurious Correlation

During fine-tuning, the model learned a false association: "If news mentions a lawsuit, it must be C12." It fails to distinguish between core and background news.

Global Intent Deficit

Despite the 512-token context, DistilBERT still lacks the complex reasoning to separate the "Overall Strategy" from the "Legal Environment."

Lessons Learned

1 Context is King

Expanding *max_length* (256 $\rightarrow$ 512) was the most significant boost, proving news summaries often hide critical cues in closing paragraphs.

2 Threshold Engineering

The default 0.5 boundary is rarely optimal for imbalanced data; dynamic sweeping is essential for real-world robustness.

3 Annealing Efficiency

Dropping the learning rate (3e-5 $\rightarrow$ 1e-5) in Stage-2 is more resource-efficient than extended training, providing a "fine-polish" to model weights.

4 Engineering Integrity

A strong baseline (optimized LR) is the only way to prove the true ROI of Deep Learning.



Possible Improvements

Architecture & Loss

Implement Focal Loss to penalize errors on "easy" samples and force the model to learn "hard" rare labels.

Upgrade to RoBERTa-Large for better logical reasoning in ambiguous cases (e.g., Strategy vs. Legal).

Data & Scaling

Unlock the full 300k corpus to increase the exposure of rare labels during training.

Explore Zero-shot prompting with Large Language Models (LLMs) to handle extreme long-tail categories with zero training samples.

/Text Analysis difficulty assessment



Difficulty Assessment

Linyao & Aojie

Data Challenges



Ordered but Fuzzy

CEFR labels are ordered but fuzzy



Sparse Intermediate Levels

Intermediate levels such as **1.5** and **2.5** are sparse



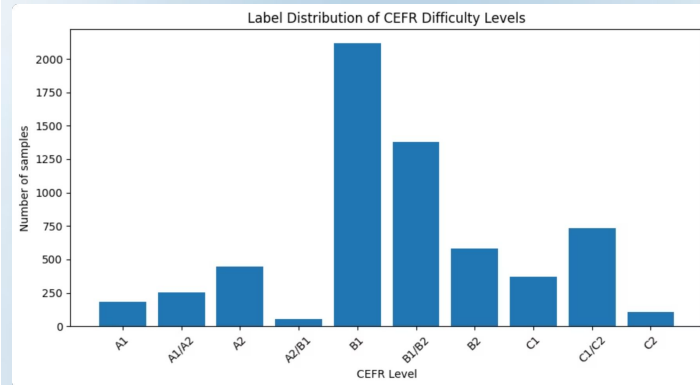
Multiple Factors

Text difficulty depends on multiple fact

Two feature versions existed:

- full feature set: 400+ features
- cleaned feature set: 114 columns

Final model used the **114-feature version** because it matched the official test set and outperformed the full feature set in later model experiments.



Baselines & Feature Selection

O1

Ridge Regression as linear baseline

O2

XGBoost Regression as nonlinear feature baseline

O3

XGBoost was also used for feature selection

In the reduced feature set, after preprocessing, **97 usable features** remained

XGBoost feature-selection ablation showed **Top 60 features** performed best overall.

Model	MAE	RMSE	R ²	Rounded Acc.
XGBoost Top 50	0.5184	0.7524	0.5758	0.4413
XGBoost Top 60	0.5179	0.7509	0.5774	0.4510
XGBoost All	0.5210	0.7511	0.5772	0.4413

From BERT Regression to Ordinal Modeling



Ordinal

BCEWithLogitsLoss



Regression

SmoothL1Loss

Model Stage	Main Idea	Observation
BERT text-only regression	text representation only	strong neural baseline
BERT + features regression	add linguistic signals	features improved prediction
BERT + features ordinal	model ordered labels	better exact CEFR prediction

Model	MAE	RMSE	R ²	Accuracy
BERT text-only regression	0.3797	0.6002	0.73	59.57%
BERT + top100 features regression	0.3499	0.5707	0.7559	62.86%
BERT + top100 features ordinal	0.2709	0.6276	0.7048	75.80%

Synthetic Data Augmentation



Source

UniversalCEFR English CEFR-labeled data



Translation Model

Helsinki-NLP OPUS-MT en-fi



Filtering

- remove empty translations
- remove duplicates
- length-ratio filtering



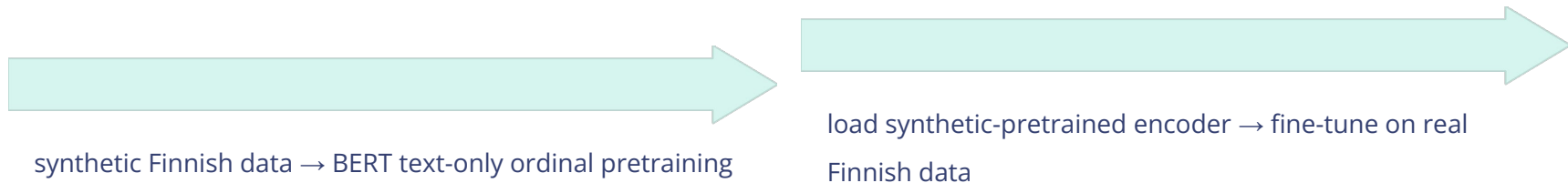
Result

Final synthetic data: **2,118 Finnish translated samples**

Synthetic Data Distribution

CEFR Level	Numeric Label	Count	Percentage
A1	1.0	123	5.81%
A2	2.0	421	19.88%
B1	3.0	436	20.59%
B2	4.0	457	21.58%
C1	5.0	456	21.53%
C2	6.0	225	10.62%
Total	—	2,118	100.00%

Training Strategy



Model Comparison Results

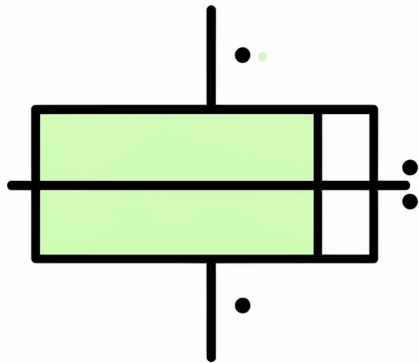
Model	MAE	RMSE	R ²	Accuracy
BERT + Top100 Features Ordinal	0.2709	0.6276	0.7048	75.80%
Synthetic-pretrained BERT + Top100 Features Ordinal	0.2548	0.5998	0.7304	77.49%

Final Results

Model	MAE	RMSE	R ²	Accuracy
Ordinal Hard	0.2251	0.5883	0.7406	80.71%
Ordinal Probability Expected Score + Rounding	0.2267	0.5811	0.7470	79.50%
Calibrated Ordinal Threshold 0.36	0.2215	0.5791	0.7486	81.03%
Regression Score + Rounding	0.2428	0.5599	0.7650	74.44%

Test result from submitted model (Ordinal Hard):

- MSE: 0.3320
- RMSE: 0.5762
- MAE: 0.2420
- R²: 0.7518
- Exact Accuracy: 78.21%
- Within ±1.0: 94.94%



Error Analysis

Strong Performance

Model performs very well on **3.0, 3.5, and 5.5**

Error Concentration

Errors concentrate on: 1.5, 2.5, 4.0, 5.0

RMSE Drivers

RMSE is mainly driven by a small number of large-error cases

Dataset

Ambiguity Similar output patterns across models suggest dataset-level ambiguity

Lessons Learned



More features are not always better



The reduced 114-feature set was cleaner and more test-compatible



Synthetic pretraining improved the BERT encoder



Ordinal classification is better for exact CEFR labels



Regression is better for RMSE/R²

Possible Improvements

- 1** better threshold calibration
- 2** class-weighted training for weak labels
- 3** higher-quality Finnish synthetic data
- 4** inspect ambiguous CEFR boundary samples manually